

Kaya Omer

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EDUCATION & AWARDS

University of Toronto

Bachelor of Science in Computer Science, Focus in Artificial Intelligence

Toronto, ON

Sep. 2023 – May 2027

President's Scholar of Excellence

Awarded to the most highly qualified undergraduate students.

Associated with University of Toronto

March 2023

Janet Faulds and M. Ann Marshall Family Award.

Awarded to the qualified students at St Michael's College.

St. Michael's College at University of Toronto

Nov 2023

EXPERIENCE

Agentic AI Researcher Intern

May 2026 – Present

The Matter Lab

Toronto, ON

- Engineering specialized AI agents for **quantum chemistry** workflows using **PySCF** and **ORCA** to automate electronic structure calculations, molecular property computation, and API interfacing.
- Architecting autonomous LLM-driven scientific discovery pipelines ("**El Agente**"), building agentic workflows that orchestrate quantum chemistry tools, external APIs, and data retrieval systems.
- Developing an AI-native framework integrating LLM-based agents with internal web interfaces, enabling automated execution of complex computational chemistry protocols.

Machine Learning Research Assistant

April 2026 – August 2026

CECCS

Toronto, ON

- Modernized legacy ML classification pipeline using **Python**, **scikit-learn**, and **NLP** (word2vec, embeddings), retraining **Random Forest** models to fix silent feature-loss bugs.
- Architected fault-tolerant, automated **LLM evaluation & inference pipelines** featuring structured output enforcement (JSON schema), exponential back-off, and cross-model agreement validation, reducing target false positives by 4x.
- Engineered **ETL pipelines (Pandas, Polars)** to process and deduplicate 20,000+ raw records; applied **statistical calibration** to optimize model thresholds and cut over-matching errors.

Software Engineer Intern

Jan 2026 – April 2026

CIBC

Toronto, ON

- Engineered internal AI integrations to automate workflows; built data pipelines using Python, Apache Spark, and Databricks to process large-scale financial data.
- Implemented MLOps workflows using Azure AI Studio, Azure ML, and Azure DevOps, including CI/CD pipelines for automated model deployment.

PROJECTS

Quantum Chemistry Agent Pipeline | *Python, PySCF, ORCA, LLMs*

May 2026

- Built LLM-driven agentic workflows to automate quantum-chemistry calculations via PySCF and ORCA, orchestrating multi-step electronic structure computations with tool-use and automated error handling.

3D Reconstruction & Automation Pipeline for SDL | *Python, OpenCV, NumPy, RPi*

Sep 2025

- Engineered a CV pipeline at Acceleration Consortium for a Self-Driving Lab combining SfM-based 3D reconstruction with automated robotic scanning for closed-loop material quality assessment.

KnowBody | *JavaScript, Cloudflare Workers, React, Workers AI (Llama 3.3)*

Jan 2026

- Built a research-based health platform using React and ES6+ JavaScript with real-time ML-driven grading and state management across complex daily metric inputs.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, Swift, JavaScript, HTML/CSS, R, Assembly

Frameworks: PyTorch, TensorFlow, React, Flask, FastAPI

Developer Tools: Git, Docker, CUDA, GCP, Azure AI Studio, Databricks, Linux

Libraries: PySCF, SciPy, OpenCV, pandas, NumPy, Matplotlib, ORCA

AI/ML: LLM Agents, Agentic Workflows, Quantum Chemistry Automation, MLOps, Computer Vision